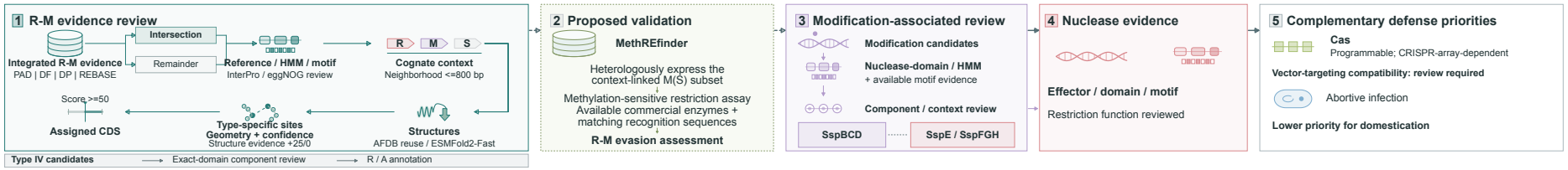
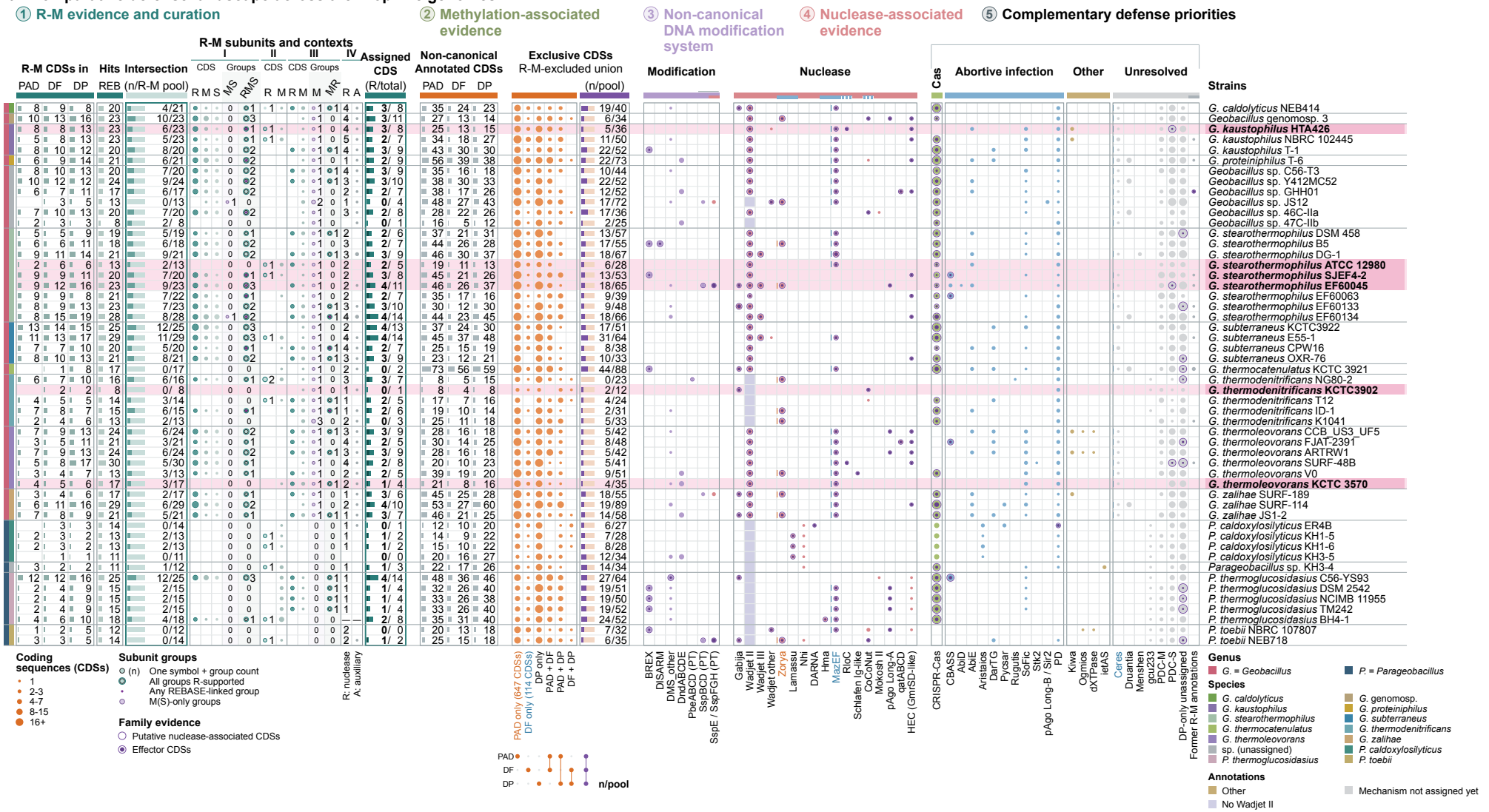
a Evidence criteria for DNMB-guided barrier prioritization



b Comparative defense landscape across thermophilic genomes



c CDS inclusion by assigned R-M type

R(n): incomplete context · M/S(n): included orphans · —: unavailable or not applicable

G. stearothermophilus EF60045

Type	Pool	Shared-pool CDSs DF	PAD	DP	DNMB	DNMB components R	M	S	A
I	9	9	9	9	9	3	3	3	—
II	0	0	0	0	0	0	0	—	—
III	3	2	2	2	2	1 (1)	1 (1)	—	—
IV	3	1	1	3	3	2	—	—	1
Total	**15**	**12**	**12**	**14**	**14**	**6 (1)**	**4 (1)**	**3**	**1**

G. stearothermophilus SJEF4-2

Type	Pool	Shared-pool CDSs DF	PAD	DP	DNMB	DNMB components R	M	S	A
I	7	7	7	6	5	2 (1)	2	1	—
II	2	2	2	1	2	1	1	—	—
III	1	0	0	0	1	0	1 (1)	—	—
IV	2	0	1	1	2	2	—	—	0
Total	**12**	**9**	**10**	**8**	**10**	**5 (1)**	**4 (1)**	**1**	**0**

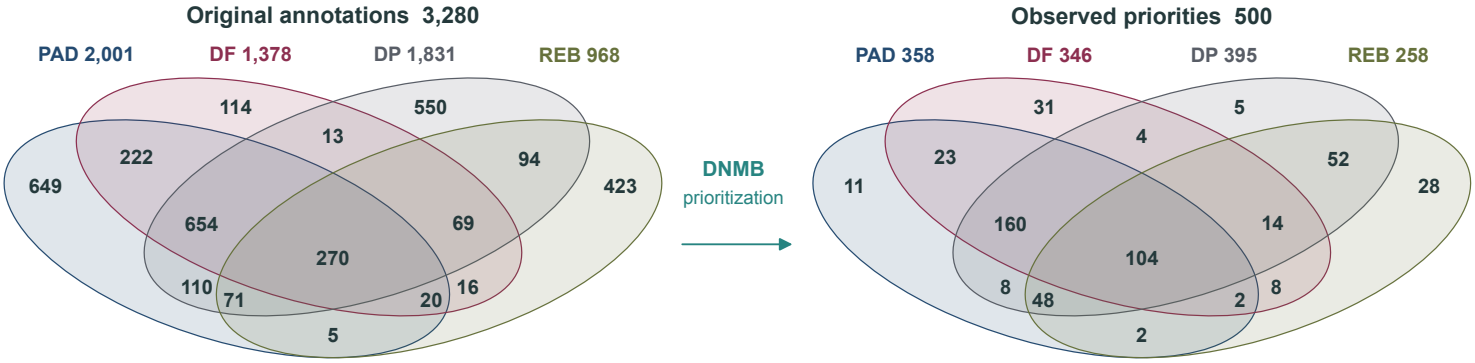
B. coagulans DSM 1

Type	Pool	Shared-pool CDSs DF	PAD	DP	DNMB	DNMB components R	M	S	A
I	6	6	6	6	6	2	2	2	—
II	1	0	0	0	1	0	1 (1)	—	—
III	2	2	2	2	2	1	1	—	—
IV	3	1	1	3	3	2	—	—	1
Total	**12**	**9**	**9**	**11**	**12**	**5**	**4 (1)**	**2**	**1**

DNMB integrated defense comparison

52 genomes · Coding sequences (CDSs)

a DNMB prioritization



Areas schematic

R subunits + nuclease effectors

b Defense categories

Category	Number of genomes	DNMB	Putative nuclease-associated CDSs		DF	PAD	DP
			Total	Effector CDSs			
R-M	52	540	422	232	365	333	425
Modification	31	127	63	10	61	127	82
Nuclease	51	352	329	132	287	283	227
CRISPR-Cas	42	501	481	104	467	500	394
Abortive infection	50	213	23	6	118	204	36
Other mechanisms	7	15	0	0	13	15	0
Unresolved	52	1,192	15	15	73	558	662
Former R-M residual	11	12	1	1	0	4	11

c Defense families

Family / pool	Number of genomes	DNMB	Putative nuclease-associated CDSs		DF	PAD	DP
			Total	Effector CDSs			
R-M (current union)	52	540	422	232	365	333	425
BREX	8	48	48	8	37	48	45
DISARM	1	5	5	1	5	5	5
DMS_other	18	26	3	1	2	26	11
DndABCDE	7	37	0	0	7	37	14
PbeABCD (PT)	1	1	1	0	0	1	1
SspBCD (PT)	4	12	6	0	12	12	8
SspE / SspFGH (PT)	4	10	6	6	10	10	10
Gabija	9	20	20	10	20	20	17
Wadjet II	33	132	132	33	131	132	130
Wadjet III	4	16	16	4	16	16	12
Wadjet other	4	6	4	2	0	6	2
Zorya	11	45	45	11	18	45	4
Lamassu	3	9	9	3	9	9	9
Nhi	6	6	0	0	0	6	6
DARNA	1	1	1	1	1	1	1
Hma	1	3	3	1	3	3	3
MazEF	29	60	60	30	60	0	0
RloC	2	2	2	2	2	2	2
Schlafen Ig-like	1	1	1	1	1	0	0
CoCoNut	7	8	4	4	8	0	7
Mokosh II	2	2	0	0	0	2	2
pAgo Long-A	8	8	8	8	8	8	3
qatABCD	2	4	4	2	0	4	0
HEC (GmrSD-like)	19	19	14	14	0	19	19
CRISPR-Cas	42	501	481	104	467	500	394
CBASS	5	24	22	5	24	24	15
AbiD	1	1	0	0	1	1	1
AbiE	16	32	0	0	32	32	0
Aristaios	4	6	0	0	6	3	6
DarTG	11	22	0	0	22	20	9
Pycsar	2	4	0	0	2	2	1
Rugutis	1	1	0	0	1	0	1
SoFic	31	43	0	0	26	43	0
Stk2	1	1	0	0	1	0	0
pAgo Long-B / Sir2	1	2	0	0	2	2	2
PD	46	77	1	1	1	77	1
Kiwa	5	10	0	0	10	10	0
Ogmios	3	3	0	0	3	3	0
dXTPase	3	3	0	0	3	3	0
ietAS	1	2	0	0	0	2	0
Ceres	28	33	0	0	33	0	0
Druantia	6	26	0	0	20	26	21
Menshen	1	2	0	0	0	2	2
gcu233	15	15	0	0	15	0	0
PDC-M	40	122	0	0	0	122	33
PDC-S	52	408	3	3	5	408	20
DP-only unassigned	52	586	12	12	0	0	586
Former R-M annotations	11	12	1	1	0	4	11

Putative nuclease-associated CDSs · Effector CDSs · Rows overlap

Counts: recorded annotations · Track: row union · —: unavailable

PAD: PADLOC; DF: DefenseFinder; DP: DefensePredictor; REB: REBASE